

Name: _____

Date: _____

VECTORS, (WRITING AND REPRESENTING VECTORS)

LO: I can write vectors in column form and represent them on a grid.

Representing vectors

A **vector** represents a movement with both **magnitude** (size) and **direction**. Written as a column vector (x, y) or using letters like **a** or AB with an arrow.

On a grid: The top number is horizontal movement (+ right, - left). The bottom is vertical (+ up, - down).

Quick examples

●	Right 3, up 4 = $(3, 4)$	positive both
●	Left 2, down 5 = $(-2, -5)$	negative both
●	$ a = (x^2 + y^2)$	magnitude formula
●	Left 3 = $(3, 0)$	left is negative: $(-3, 0)$
●	Vectors must start at the origin	can start anywhere

Key idea

Column vector (x, y) : x = horizontal, y = vertical. Magnitude = $(x^2 + y^2)$.

$$|(x, y)| = (x^2 + y^2)$$

Use Pythagoras to find the magnitude

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - writing from grid

A point moves right 4 and up 3. Write as a column vector.

Horizontal: +4 (right)

Vertical: +3 (up)

Vector = $(4, 3)$

$(4, 3)$

Right and up are positive.

Example 2 - negative components

A point moves left 2 and down 6. Write as a column vector.

Horizontal: -2 (left)

Vertical: -6 (down)

Vector = $(-2, -6)$

$(-2, -6)$

Left and down are negative.

Example 3 - finding magnitude

Find the magnitude of $(3, 4)$.

$$\begin{aligned} |a| &= (3^2 + 4^2) \\ &= (9 + 16) \\ &= 25 = 5 \end{aligned}$$

$|a| = 5$

Use Pythagoras: $(x^2 + y^2)$.

Example 4 - from coordinates

A = $(1, 2)$, B = $(5, 5)$. Find vector AB.

$$\begin{aligned} AB &= B - A \\ &= (5-1, 5-2) \\ &= (4, 3) \end{aligned}$$

$AB = (4, 3)$

Subtract start from end.

YOUR TURN – PRACTICE (deliberate practice)

1. Writing vectors from movement

Write each movement as a column vector.

- a) Right 5, up 2
- b) Left 3, up 7
- c) Right 1, down 4
- d) Left 6, down 3

2. From coordinates

Find the vector from A to B.

- a) A(1,3), B(4,7)
- b) A(5,2), B(1,6)
- c) A(0,0), B(3,-2)
- d) A(7,4), B(2,1)

3. Finding magnitude

Find the magnitude of each vector.

- a) (3, 4)
- b) (5, 12)
- c) (6, 8)
- d) (-4, 3)

4. Equal vectors

State which vectors are equal.

- a) (3,2) from (0,0) and (3,2) from (1,5)
- b) (4,1) and (1,4)
- c) AB where A(1,2) B(4,5) and CD where C(0,1) D(3,4)
- d) (5,0) and (0,5)

5. Mixed problems

Solve each.

- a) Vector (a, 3) has magnitude 5. Find a.
- b) AB = (4, -3). If A = (2, 5), find B.
- c) Magnitude of (1, 1)
- d) Vector from (3, 7) to (3, 2)
- e) Unit vector in direction (3, 4)

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) A column vector (x, y) means right x and up y ☐

Correct answer: _____

Reason: _____

b) Vectors must start at the origin ☐

Correct answer: _____

Reason: _____

c) Magnitude = $(x^2 + y^2)$ ☐

Correct answer: _____

Reason: _____

d) Left 3 is written as (3, 0) ☐

Correct answer: _____

Reason: _____

e) Equal vectors have the same magnitude AND direction ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A vector has magnitude **10** and makes an angle of **30°** with the positive x-axis. Write it as a column vector using exact trig values.

Expression: _____

Simplified: _____

b) Points A(2,1), B(6,4), C(10,7). Show that A, B, C are collinear and find the ratio AB:BC.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

